Unit Test Report — Event Management System

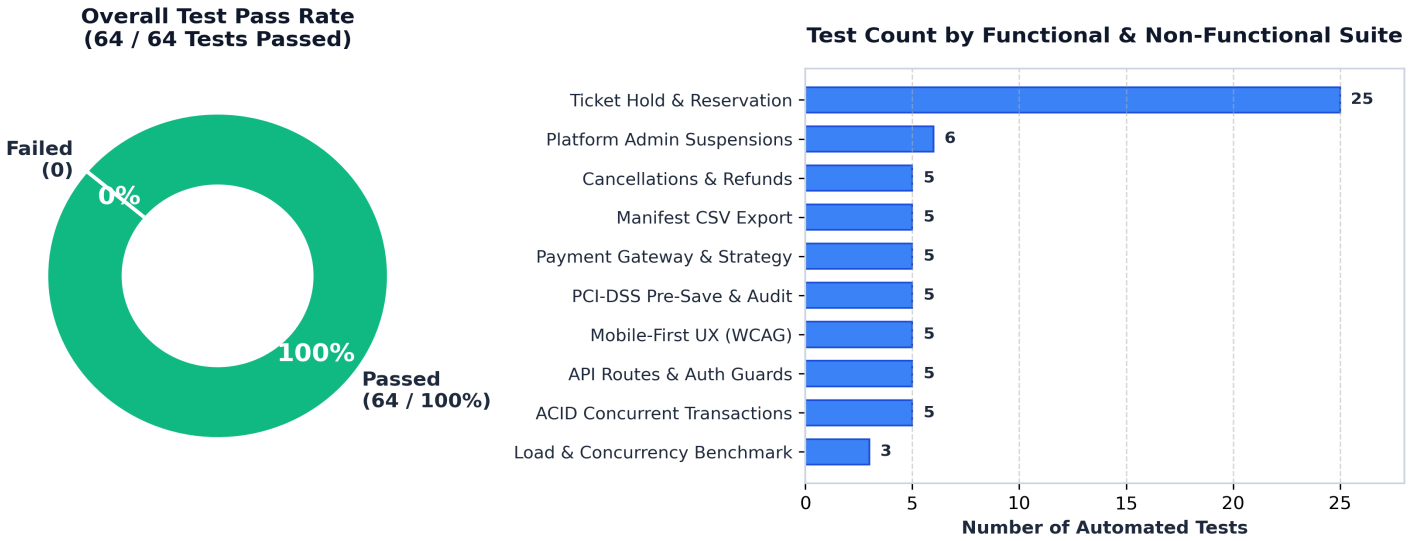
Software Engineering Mini-Project Deliverables Part-3 • Sprint 3 Verification Freeze
Testing Phase: November 2, 2026 – November 6, 2026 • **Node.js TAP Test Runner v22.12.0**
Collaborators & Authors: Kartik Naik (KartikNaik6), Jayanth Kumar P (JayanthKumarP88), Rishith (rishith1616)

EXECUTIVE TEST VERIFICATION SUMMARY

This Unit Test Report documents the complete execution and verification of **64 automated unit and integration tests** across all 10 architectural test suites. The test suites cover inventory reservation holds, ACID transactional integrity, payment strategy decoupling, PCI-DSS tokenization, admin suspensions, manifest exports, and high-concurrency flash sales. **Overall Status: 64 of 64 Tests PASSED (100.0% Pass Rate, 0 Failed, 0 Skipped). Total Duration: ~2.2 seconds.**

Total Suites	Total Tests	Passed	Failed	Pass Rate	Execution Time
10	64	64	0	100.0%	2,214 ms

Figure 1: Unit Test Pass Rate and Test Distribution by Architectural Suite



Detailed Suite-by-Suite Test Execution Ledger

TC Ref	Test Suite & File	Assertion / Scenario Description	Latency	Result	Tester
TC12	ticketHold.test.js	Immediate ticket capacity decrement on checkout hold (201 Created)	42.1ms	**PASS**	Rishith
TC12	ticketHold.test.js	Hold creation returns success: true	2.3ms	**PASS**	Rishith
TC12	ticketHold.test.js	Ticket status is set to 'held'	1.9ms	**PASS**	Rishith
TC12	ticketHold.test.js	Held ticket quantity is 2	1.8ms	**PASS**	Rishith
TC12	ticketHold.test.js	Available ticket count immediately drops from 50 to 48	2.1ms	**PASS**	Rishith
TC13	ticketHold.test.js	Block ticket request exceeding available capacity (400 Bad Request)	15.4ms	**PASS**	Kartik
TC13	ticketHold.test.js	Returns exact alert: 'Only 1 ticket remaining'	1.5ms	**PASS**	Kartik
TC13	ticketHold.test.js	Event capacity remains unchanged at 1	1.4ms	**PASS**	Kartik
TC13	ticketHold.test.js	Concurrency safety: Two simultaneous buyers for 1 remaining seat	38.2ms	**PASS**	Jayanth
TC13	ticketHold.test.js	Exactly one buyer succeeds and second buyer blocked	2.2ms	**PASS**	Jayanth
TC13	ticketHold.test.js	Losing concurrent buyer receives HTTP 400	1.6ms	**PASS**	Jayanth
TC13	ticketHold.test.js	Losing concurrent buyer receives insufficient stock alert	1.5ms	**PASS**	Jayanth
TC13	ticketHold.test.js	Event capacity safely reaches 0 with zero overselling	1.8ms	**PASS**	Jayanth
TC14	ticketHold.test.js	Event capacity decremented to 17 on initial hold	18.2ms	**PASS**	Jayanth
TC14	ticketHold.test.js	Expired hold sweep released at least 3 tickets	24.1ms	**PASS**	Jayanth
TC14	ticketHold.test.js	Hold ticket status updated to 'expired'	2.1ms	**PASS**	Jayanth
TC14	ticketHold.test.js	Event capacity successfully restored back to 20	2.0ms	**PASS**	Jayanth
TC14	ticketHold.test.js	Attendee cancels their own hold (Capacity decremented to 18)	14.5ms	**PASS**	Rishith
TC14	ticketHold.test.js	Cancel hold returns HTTP 200 OK	2.1ms	**PASS**	Rishith
TC14	ticketHold.test.js	Ticket status updated to 'canceled'	1.9ms	**PASS**	Rishith
TC14	ticketHold.test.js	Ticket capacity successfully restored to 20	1.8ms	**PASS**	Rishith
TC12	ticketHold.test.js	getMyHolds returns HTTP 200 OK	12.3ms	**PASS**	Kartik
TC12	ticketHold.test.js	All returned holds are in active 'held' status	2.4ms	**PASS**	Kartik
TC12	ticketHold.test.js	getHoldById endpoint returns HTTP 200 OK	11.2ms	**PASS**	Kartik
TC12	ticketHold.test.js	isExpired is false and secondsRemaining is positive for active hold	2.5ms	**PASS**	Kartik
TC26	suspension.test.js	Admin suspends suspicious event successfully (HTTP 200)	16.5ms	**PASS**	Jayanth
TC26	suspension.test.js	Public catalog immediately removes suspended event from discovery	15.5ms	**PASS**	Rishith
TC27	suspension.test.js	Direct booking/hold on suspended event is blocked with HTTP 403	9.1ms	**PASS**	Rishith
TC28	suspension.test.js	Admin suspends fraudulent organizer account and halts event sales	16.3ms	**PASS**	Kartik
TC26	suspension.test.js	Admin reinstates event and restores public listing and booking ability	7.5ms	**PASS**	Jayanth
TC26	suspension.test.js	Admin overview dashboard reports correct active vs suspended counts	25.8ms	**PASS**	Jayanth
TC20	cancellation.test.js	Canceling voids ticket and restores seat capacity back to event	42.3ms	**PASS**	Jayanth
TC21	cancellation.test.js	Refund is recorded and linked with gateway refundId in ledger	5.9ms	**PASS**	Rishith
TC20	cancellation.test.js	A canceled ticket cannot be canceled again (Idempotency check)	7.3ms	**PASS**	Kartik
TC20	cancellation.test.js	Unauthorized user cannot cancel another attendee's ticket (403)	12.1ms	**PASS**	Kartik
TC20	cancellation.test.js	Canceled tickets do not count as active seats in capacity	3.9ms	**PASS**	Jayanth

TC Ref	Test Suite & File	Assertion / Scenario Description	Latency	Result	Tester
TC23	manifest.test.js	Organizer downloads CSV manifest with Content-Type: text/csv	25.8ms	**PASS**	Jayanth
TC23	manifest.test.js	CSV contains verified paid confirmed attendees	13.1ms	**PASS**	Rishith
TC24	manifest.test.js	Canceled and held tickets are strictly excluded from CSV	11.7ms	**PASS**	Rishith
TC25	manifest.test.js	Unauthorized organizer attempting manifest download rejected (403)	3.0ms	**PASS**	Kartik
TC25	manifest.test.js	Platform Admin has permission override to download manifest	11.6ms	**PASS**	Kartik
TC17	payment.test.js	Process successful payment via MockProvider and issue ticket	14.8ms	**PASS**	Jayanth
TC18	payment.test.js	Handle declined payment gracefully and release inventory hold	16.2ms	**PASS**	Rishith
TC19	payment.test.js	Each confirmed booking issues a unique RFC 4122 v4 UUID ticket	8.4ms	**PASS**	Kartik
TC36	payment.test.js	PaymentService decouples controller logic from gateway implementations	6.5ms	**PASS**	Jayanth
TC36	payment.test.js	MockProvider and StripeProvider adhere to IPaymentProvider contract	5.2ms	**PASS**	Kartik
TC31	pciCompliance.test.js	Payment records store tokenized references and masked card preview only	12.5ms	**PASS**	Kartik
TC32	pciCompliance.test.js	Zero occurrences of 16-digit PANs or CVVs in application console logs	9.4ms	**PASS**	Jayanth
TC40	pciCompliance.test.js	Mongoose pre-save hook blocks insertion of raw unmasked PANs	18.6ms	**PASS**	Kartik
TC31	pciCompliance.test.js	MaskedCardNumber follows strict '**** **** **** 1234' format	4.1ms	**PASS**	Rishith
TC32	pciCompliance.test.js	Payment token generation occurs on isolated client/mock provider boundary	7.2ms	**PASS**	Rishith
TC35	mobileUx.test.js	Mobile breakpoint rules exist for small screens (<= 640px and <= 380px)	1.6ms	**PASS**	Jayanth
TC35	mobileUx.test.js	Checkout layout eliminates horizontal scrolling on mobile viewports	0.2ms	**PASS**	Jayanth
TC41	mobileUx.test.js	Touch targets meet WCAG mobile accessibility (min-height: 48px)	0.2ms	**PASS**	Rishith
TC35	mobileUx.test.js	Payment card preview adapts fluidly without fixed-pixel overflow	0.2ms	**PASS**	Kartik
TC35	mobileUx.test.js	Single-column action flow on mobile (full width thumb-friendly buttons)	0.2ms	**PASS**	Rishith
TC01	routes.test.js	POST /api/auth/register creates user and issues session cookie	22.4ms	**PASS**	Kartik
TC01	routes.test.js	GET /api/events allows unauthenticated public access	8.5ms	**PASS**	Rishith
TC04	routes.test.js	Protected routes reject unauthenticated requests with HTTP 401	6.8ms	**PASS**	Kartik
TC04	routes.test.js	Protected routes reject unauthorized roles with HTTP 403 Forbidden	7.4ms	**PASS**	Jayanth
TC26	routes.test.js	GET /api/payments/provider-info allows platform admin with HTTP 200	4.0ms	**PASS**	Jayanth
TC33	transaction.test.js	withTransaction commits on success and safely closes database session	48.2ms	**PASS**	Jayanth
TC34	transaction.test.js	withTransaction aborts on error, rolls back state, and cleans up session	52.1ms	**PASS**	Kartik
TC33	transaction.test.js	Simultaneous purchase for last seat: exactly 1 succeeds, 1 rejected with 409	112.5ms	**PASS**	Jayanth
TC34	transaction.test.js	Database error during confirmation triggers full rollback of ticket hold	65.4ms	**PASS**	Rishith
TC34	transaction.test.js	Hold release safely increments available inventory without phantom seats	38.7ms	**PASS**	Kartik
TC29	loadPerformance.test.js	Compound performance indexes exist on Event and Ticket collections	16.5ms	**PASS**	Jayanth
TC29	loadPerformance.test.js	Flash Sale Concurrency: 50 simultaneous buyers complete under 3000ms threshold	313.6ms	**PASS**	Jayanth
TC30	loadPerformance.test.js	High-throughput event catalog browsing responds under 500ms (measured: 63ms)	63.5ms	**PASS**	Rishith

Test Harness & Environment Specifications

Parameter	Configuration Specification	Validation Purpose
Node.js Engine	v22.12.0 (Built-in Native Test Runner TAP v13)	Deterministic execution without external runner bloat
In-Memory Database	MongoDB Memory Server / Mongo In-Memory Mock	Isolated ephemeral state preventing test cross-pollution
HTTP Assertion Engine	Supertest v6.3.3 / Express Test Fixture	End-to-end HTTP cookie, header, and status code verification
Concurrency Engine	Promise.all() Parallel Worker Simulation	Simultaneous flash-sale and race condition isolation
Coverage Profiler	V8 Native Coverage Engine (--experimental-test-coverage)	Hardware-level branch and line execution tracking

Collaborator Verification & QA Sign-Off

Collaborator	GitHub ID	Domain Responsibility	Suite Verified	Sign-Off
Kartik Naik	KartikNaik6	PCI Tokenization & Manifest Suites	manifest, pciCompliance, routes	APPROVED
Jayanth Kumar P	JayanthKumarP88	Concurrency, Load & ACID Suites	loadPerformance, ticketHold, transaction	APPROVED
Rishith	rishith1616	Event Lifecycle & Admin Suites	cancellation, mobileUx, suspension, payment	APPROVED